

Nguyen Tan Loc

Software Engineer

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Address: An Khanh, Ninh Kieu, Can Tho

ABOUT

Software Engineer with experience in backend and frontend web development, specializing in scalable APIs, transaction workflows, real-time applications, and business management systems. Proficient in NestJS, Go, React, PostgreSQL, and MongoDB, with a strong focus on writing clean, maintainable, and reliable software.

SKILLS

- *Programming languages:* Javascript, TypeScript, Go, SQL, NoSQL.
- *Frameworks and libraries:* NodeJS(NestJS), React, Ant Design, Sass, CSS.
- *Databases:* PostgreSQL, MySQL, MongoDB.
- *Programming paradigms and architectures:* RESTful API, GraphQL, Monolith, Microservices.
- *Soft skills:* Teamwork, responsibility, problem solving, adaptive.

EXPERIENCE

- **TEKNIX TECHNOLOGY CORPORATION**
Backend Developer - NestJS & Go(Centralized Wallet & Social Chat)
March 2025 - March 2026
Ninh Kieu, Can Tho

- Centralized Wallet: Backend developer for centralized wallet operations supporting real-time transaction on multiple networks
 - **Technologies:** NestJS, PostgreSQL, CockroachDB, Redis, RabbitMQ, Ably, Clickhouse, OTel, API Gateway(KrakenD).
 - **Responsibility:**
 - Wallet management and balance tracking across multiple blockchains.
 - Implemented real-time transaction processing, message queue consumers, rollback mechanisms, and notification handlers.
 - Implemented a hybrid security mechanism using Ed25519 signatures and AES-GCM encryption.
 - Developed user and system settings dashboards using Directus.
 - Developed and integrated an observability platform for monitoring, log analysis, tracing, and application security.
 - Designed APIs for dashboard.
 - Real-time Chat & Social Platform: Developed core social media features.
 - **Technologies:** Go(Gin), MongoDB, Peertube, R2 Cloudflare, NATs, API Gateway(KrakenD).
 - **Responsibility:**
 - Developed features for posts, reels, comments, and playlists, enabling efficient content creation, management, and user interaction.
 - Developed scheduling features for posts and reels.
 - Implemented PeerTube integration for reel video upload, management, and delivery.
 - Implemented approval workflows for posts and reels.
 - Integrated real-time push notifications to improve user engagement.
 - Designed APIs for dashboard.
- **DANG AN TECH COMPANY LIMITED (DA.Tech)**
Frontend Developer - ReactJS, CouchDB, Ant Design, HTML, SASS(POS System)
Apr 2024 - March 2025
Ninh Kieu, Can Tho

- Integrated electronic invoices from suppliers(Viettel and Mobifone)
 - Collaborated with teammates to develop and convert the frontend side

PROJECTS

- **Nest Admin Dashboard**
April 2026 – Present

Description: Developed an attribute-based access control (ABAC) system following Domain-Driven Design (DDD) principles, delivering a scalable and flexible authorization solution for modern multi-tenant applications.

Responsibility:

 - Designed and developed the administrative dashboard for managing users, roles, policies, and tenant configurations.
 - Implemented backend services and RESTful APIs following Domain-Driven Design (DDD) principles.
 - Built an attribute-based authorization engine using JsonLogicJS for dynamic policy evaluation.
 - Integrated PostgreSQL, Redis, RabbitMQ, and KrakenD to support a scalable, event-driven system architecture.
 - Designed database schemas and data access layers with Prisma.
 - Documented and maintained APIs using Swagger.

Technologies: NestJS, PostgreSQL, Swagger, Prisma, KrakenD, Redis, RabbitMQ, JsonLogicJS.

Demo: <https://cjoool-admin-dashboard.vercel.app> **Github:** <https://github.com/locnguyen2k2/laboratory-management-system>.
- **Quizzes Exams Management System**
Aug 2024 – Present

Description: Developed a question management and automatic exam paper generation system for the university that aims to helps lecturers easily personalize the generation exams and management questions, and the data related to exam papers.

Responsibility: Build the frontend-side, and server-side functionality, including randomly generating and managing exam papers, managing and classifying questions, classes, courses, chapters, users, permissions, and authorization(RBAC).

Technologies: NestJS, MongoDB, Apollo GraphQL, NodeMailer, Pinata, NextJS, Ant design, CouchDB.

Demo: <https://cjoool117-ems-system.vercel.app> **Github:** <https://github.com/locnguyen2k2/exam-management-system>.

EDUCATION

Can Tho University of Technology

Software Engineer

Major: Software engineering

GPA: 3,47/4.0

Oct 2020 - Jan 2025
NVC, Ninh Kieu, Can Tho